11/2/26 Exp: No:- 3

Determination of voltage in circuit using Nodal Analysis:-

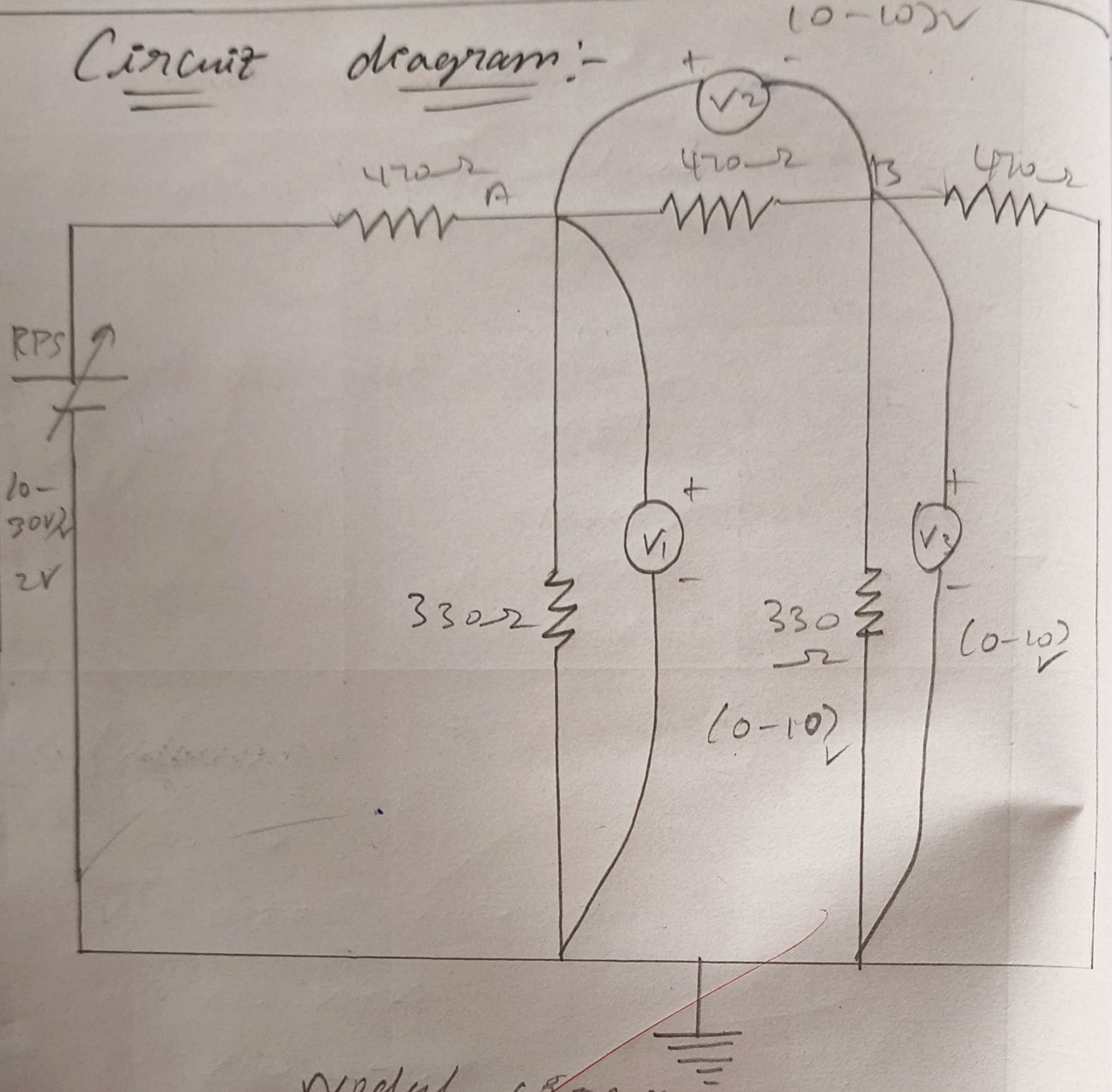
Aim:-

To determine the voltage in the circuit using nodal analysis both theoretically and Practically for a given DC circuit:-

Apparatus Required.

SL.NO	Apparatus	Specification	Quantity.
1	Regulated power supply (RPS)	(0 - 30V)	1
2	Multimeter	---	1
3	Resistors	470Ω, 330Ω	3, 2
4	Bread board	---	1

Circuit diagram:-



Nodal circuit.

Tabular column:-

Parameters	Theoretical	Practical
2V	0.92	0.98
5V	2.61	2.68
10V	5.39	5.47
14V	7.59	7.66

Verified

Calculations:-

For 2V

$V_1 = 0.20, \quad V_2 = 0.72$

$V = V_1 + V_2$

$V = 0.92V$

For 5V

$V_1 = 0.60, \quad V_2 = 2.01$

$V = V_1 + V_2$

$V = 0.60 + 2.01$

Procedure :-

1). Give connections as per the Circuit diagram.

2). Switch on the supply, vary the RPS (Regulated power supply). and set a Particular input voltage.

3). Note down the readings of ammeter and voltmeters and tabulated them.

4). Vary the RPS for different input voltages and note down the readings of all the meters.

5) Reduce the RPS to its minimum value and switch off the supply.

6). using the tabulated values, verify Kirchoff's law practically. and verify it theoretically.

$V = 2.61V$

For 10V

$V1 = 1.30, \quad V2 = 4.09$

$V = V1 + V2$

$V = 5.39V$

For 14V

$V1 = 1.81, \quad V2 = 5.78$

$V = V_1 + V_2$

$V = 1.81 + 5.78$

$V = 7.59V$

Marks Split-up:-

SL. NO	Description	Marks allotted	Marks obtained.
1	Experimental setup / connections	20 marks	20
2	Observation / Readings	20 marks	20
3	Calculations / Results	20 marks	10
4	Viva	20 marks	10
5	Record	20 marks	20
		Total	80

Result:-

Thus, the nodal analysis is verified Practically and theoretically. The resultant voltages for 2V supplies are:-

a) The voltage V_1 is 0.96 V.

b) The voltage V_2 is 2.68 V

c) The voltage V_3 is 5.47 V

d) The voltage V_4 is 7.66 V.